Phase1_analysis

March 21, 2026

[73]:
```
import importlib
import pandas as pd
import numpy as np
import re
from pathlib import Path

import functions
importlib.reload(functions)
from functions import *

df_1111 = pd.read_csv("AI_Lit_Que_1111.csv")
df_1204 = pd.read_csv("AI_Lit_Que_1204.csv")

# add source/course label
df_1111["course"] = "1111"
df_1204["course"] = "1204"

print(df_1111.shape, df_1204.shape)
```

```
(118, 37) (23, 37)
```

---

1 EDA

1.1 Descriptive statistics

1.1.1 Mean comparison

SES

- Class 1111 is very slightly below the pooled SES average
- Class 1204 is slightly above the pooled SES average

The difference is present but not large in raw standardized units.

- *Skew < 0*: SES is somewhat left-skewed, especially in 1204. More observations are clustered on the higher side, with a tail toward lower SES-index values.

AI Literacy

- Class 1204 appears to have higher AI literacy on average than class 1111. The gap here is much more visible than for SES.

[34]:
```
df, meta = prepare_dataset("AI_Lit_Que_1111.csv", "AI_Lit_Que_1204.csv")

# prepare_dataset is a preprocessing function defined in functions.py

construct_summary_table(df)
```

[34]:
```
     sample          construct    n          mean        sd    median  \
0      1111          SES index  118 -2.184274e-02  0.570420  0.003448
1      1111  AI literacy score  118  3.490678e+00  0.409291  3.500000
2      1204          SES index   23  1.120627e-01  0.569433  0.235097
3      1204  AI literacy score   23  3.800000e+00  0.450252  3.900000
4  Combined          SES index  141 -2.834612e-17  0.570388  0.020992
5  Combined  AI literacy score  141  3.541135e+00  0.430128  3.500000

        iqr      skew
0  0.792470 -0.409507
1  0.500000  0.266702
2  0.661270 -0.804122
3  0.750000  0.134169
4  0.801864 -0.460321
5  0.600000  0.294079
```

1.1.2 Measurement consistency

SES

- SES shows **moderate** consistency across both classes (pooled Cronbach's = 0.69).

AI Literacy

- AI literacy shows **low** consistency across both classes (pooled Cronbach's = 0.52; even lower in 1111).

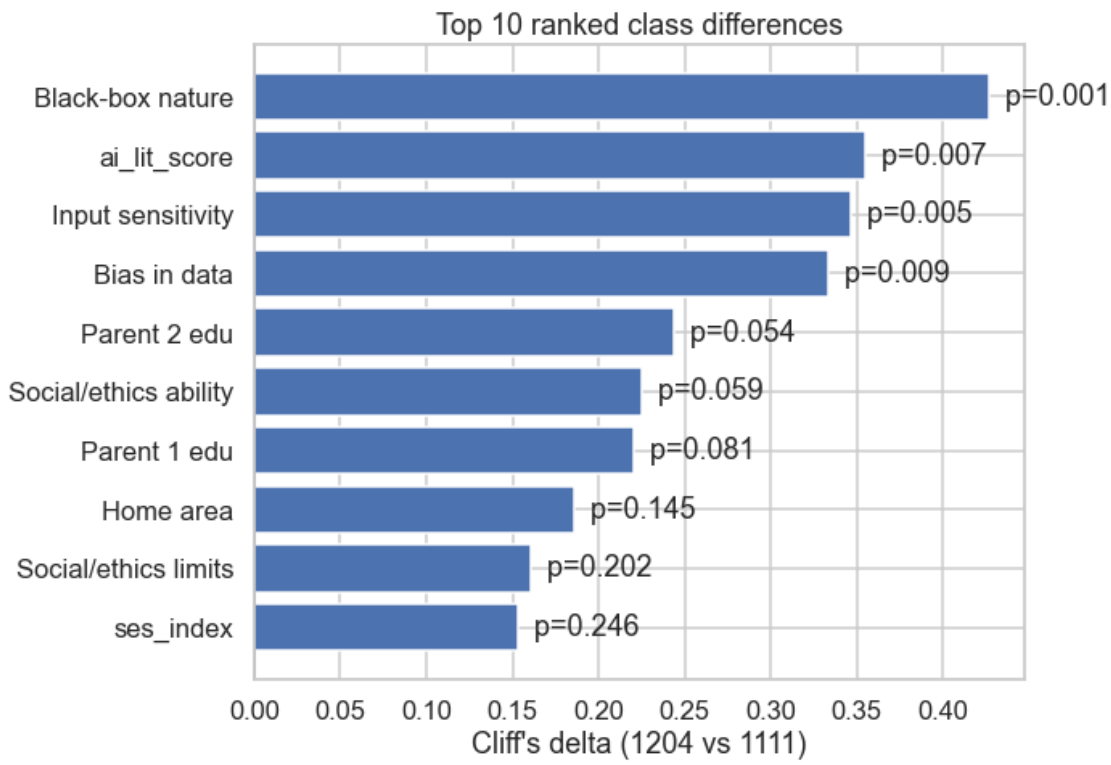
The AI literacy items do not appear to function as a highly homogeneous single scale in this sample.

[35]:
```
reliability_table(df, meta)
```

[35]:
```
     sample              construct     alpha  n_complete
0      1111  SES index ingredients  0.689986         118
1      1111      AI literacy score  0.466021         118
2      1204  SES index ingredients  0.739366          23
3      1204      AI literacy score  0.624770          23
4  Combined  SES index ingredients  0.693905         141
5  Combined      AI literacy score  0.523376         141
```

1.1.3 Effect sizes: Two-sided Mann–Whitney U test

[36]:
```
eff = effect_size_table(df, meta)
plot_ranked_effect_sizes(eff, top_n=10)
```



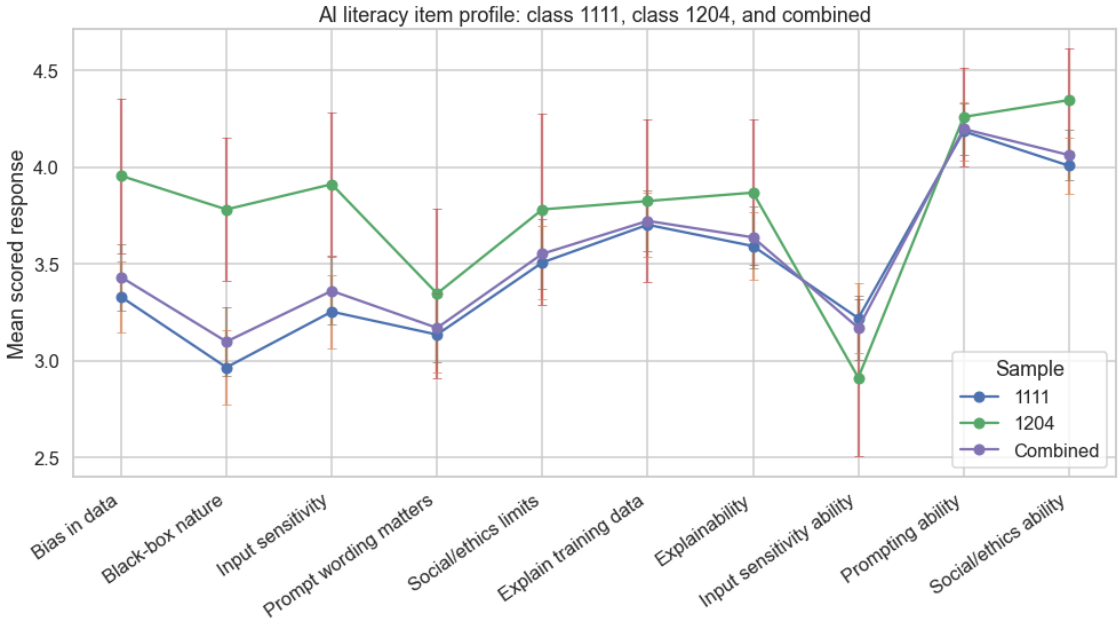
- Class 1204 tends to score higher than class 1111 on several AI literacy components, especially the more conceptual items.
- By contrast, the SES differences are weaker and mostly not statistically convincing.

1.2 Visualizations

1.2.1 Item profiles

[37]:
```
plot_item_profile(df, meta, construct="SES")
```

[38]:
```
plot_item_profile(df, meta, construct="AI")
```



1.2.2 Distribution of composite scores

[39]:
```
plot_ses_ai_correlation_heatmaps(df, meta)
```

- In the pooled sample, most correlations are weak to modest, with a tendency toward small positive associations rather than a single dominant pattern.
- The 1204 panel shows more extreme values, but given the much smaller subgroup size, those correlations are likely less stable.

---

2 SES Effect on AI Literacy (No mediation)

$$\text{AI literacy} = \beta_0 + \beta_1 \text{SES} + \varepsilon$$

[40]:
```
def run_total_effect_models(df):
    results = []

    for sample in ["1111", "1204", "Combined"]:
        res, _, _ = fit_total_effect_model(df, sample=sample)
        results.append(res)

    return pd.concat(results, ignore_index=True)

run_total_effect_models(df)
```

[40]:
```
      sample    n  beta_ses_std    se_hc3     t_hc3     p_hc3  ci_low_95  \
0       1111  118      0.194558  0.084811  2.294033  0.021789   0.028333
1       1204   23      0.109333  0.270676  0.403926  0.686267  -0.421182
2   Combined  141      0.195893  0.076050  2.575846  0.010000   0.046838

   ci_high_95  r_squared  spearman_rho  spearman_p
0    0.360784   0.037853      0.199891    0.029991
1    0.639848   0.011954      0.013161    0.952473
```

```
2    0.344948    0.038374       0.201231     0.016723
```

- In the **combined sample**, SES shows a small but statistically supported positive association with AI literacy. Students with higher SES tend to have slightly higher AI literacy scores.
- In **class 1111**, the estimated SES effect is again positive and modest, and the result is statistically significant.
- In **class 1204**, the estimated effect is also positive, but highly uncertain and not statistically supported — most likely due to the much smaller sample size rather than evidence that no relationship exists.

The effect size is not large. In both 1111 and the combined sample, a one standard deviation increase in SES is associated with only about a 0.2 SD increase in AI literacy.

The explained variance is low (R^2 0.04), meaning SES accounts for only a small part of the variation in AI literacy. SES appears to matter, but is clearly not the only factor.

- The Spearman correlations are also positive in 1111 and in the combined sample, supporting the same conclusion from a rank-based perspective.

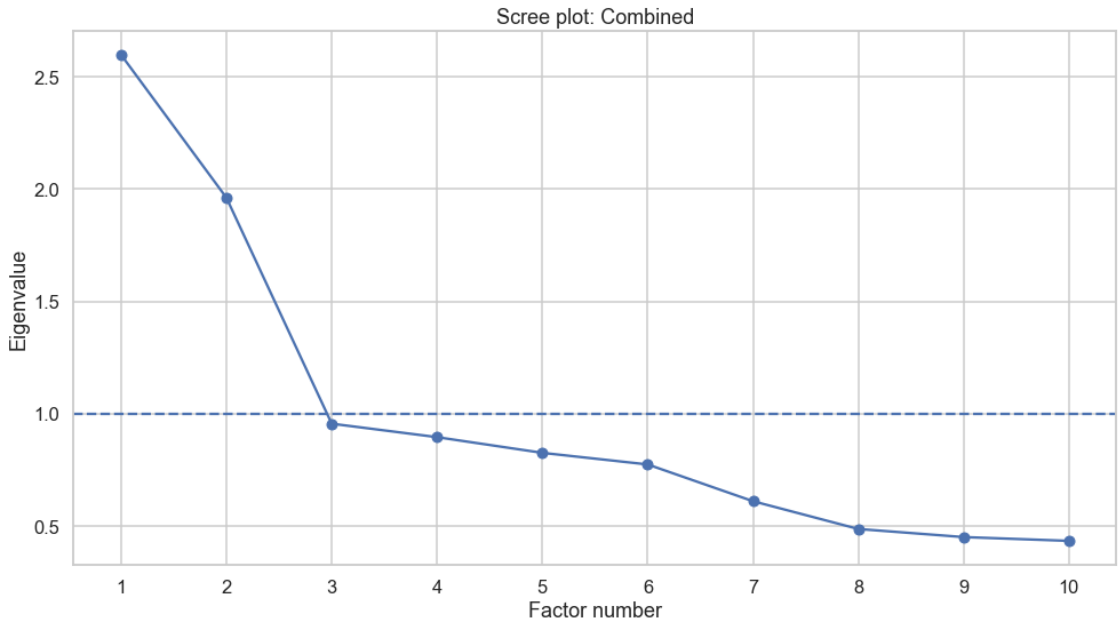
2.1 Exploratory Factor Analysis (EFA) for AI Literacy

[41]:
```
ai_efa_items = [
    "ai_concept_data_bias_scored_num",
    "ai_concept_blackbox_scored_num",
    "ai_concept_input_variation_scored_num",
    "ai_concept_prompt_wording_scored_num",
    "ai_concept_social_ethics_scored_num",
    "ai_ability_training_data_scored_num",
    "ai_ability_explainability_scored_num",
    "ai_ability_input_sensitivity_scored_num",
    "ai_ability_prompting_scored_num",
    "ai_ability_social_ethics_scored_num",
]

diag_combined, kmo_items_combined = efa_diagnostics(df, "Combined",␣
 ↪ai_efa_items)
diag_combined
```

[41]:
```
     sample  n_complete  bartlett_chi2    bartlett_p  kmo_overall
0  Combined         141     245.646175  1.921883e-29     0.703442
```

- The diagnostics suggest that running EFA on the combined AI literacy items is reasonable

[42]:
```
eig_combined = run_scree_analysis(df, "Combined", ai_efa_items)
eig_combined
```



[42]:

	factor_number	eigenvalue
0	1	2.598085
1	2	1.962121
2	3	0.956333
3	4	0.896443
4	5	0.826614
5	6	0.774900
6	7	0.611327
7	8	0.487682
8	9	0.451673
9	10	0.434821

- The eigenvalues support a two-factor modeling. 2 principal components have eigenvalues > 1.

[43]:
```
fa_combined, d_combined, loadings_combined, variance_combined = fit_efa(
    df=df,
    sample="Combined",
    items=ai_efa_items,
    n_factors=2,
    rotation="oblimin"
)

clean_loadings(loadings_combined, cutoff=0.30)
```

[43]:

	Factor1	Factor2
ai_concept_data_bias_scored_num	0.624	

```
ai_concept_blackbox_scored_num            0.598
ai_concept_input_variation_scored_num      0.44
ai_concept_prompt_wording_scored_num      0.518
ai_concept_social_ethics_scored_num        0.75
ai_ability_training_data_scored_num               0.505
ai_ability_explainability_scored_num              0.454
ai_ability_input_sensitivity_scored_num  -0.416
ai_ability_prompting_scored_num                   0.658
ai_ability_social_ethics_scored_num               0.523
```

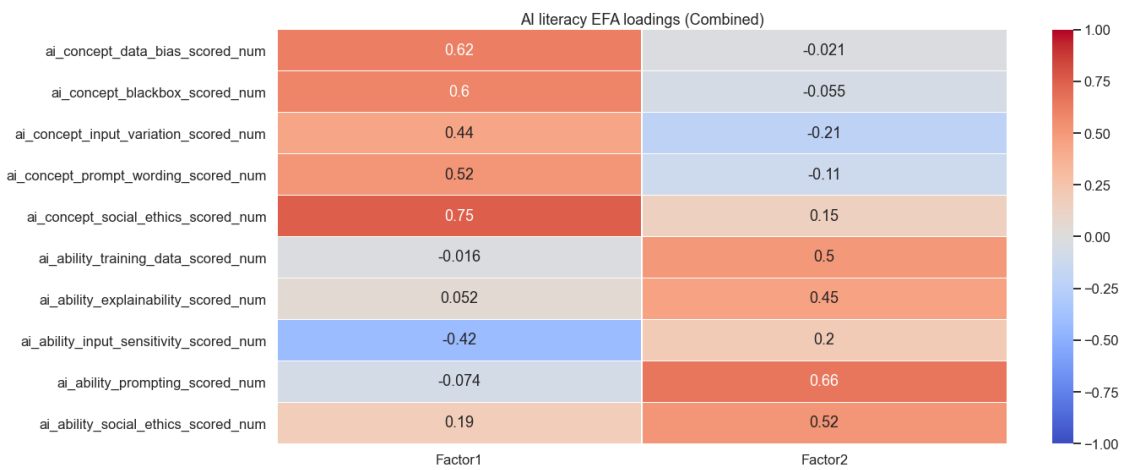
- *Factor 1 is dominated by the conceptual items*: data bias, black-box nature, input variation, prompt wording, and especially social/ethics limits all load positively on this factor.

That means Factor 1 can be interpreted as something like conceptual AI understanding

- *Factor 2 is dominated by the self-reported competence items*: explaining training data, explainability, prompting ability, and social/ethics ability load on this factor.

Factor 2 looks like self-reported AI ability/confidence.

[44]:
```
plot_loading_heatmap(
    loadings_combined,
    title="AI literacy EFA loadings (Combined)"
)
```



2.2 EFA for SES (Ordinal variables only)

[45]:
```
# some non-ordinal variables are excluded here

ses_efa_items = [
    "ses_parent1_edu_num",
    "ses_parent2_edu_num",
    "ses_household_income_num",
```

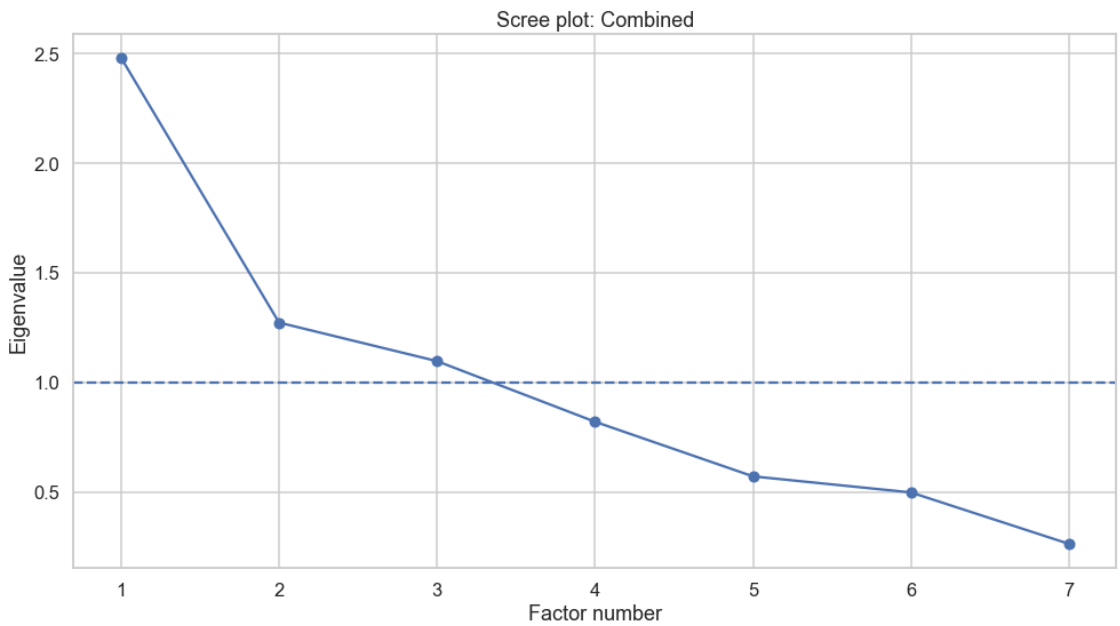
```
    "ses_home_area_num",
    "ses_device_access_scored_num",
    "ses_internet_quality_scored_num",
    "ses_financial_constraint_scored_num",
]

diag_ses_combined, kmo_ses_combined = efa_diagnostics(df, "Combined",␣
 ↪ses_efa_items)
diag_ses_combined
```

[45]:

	sample	n_complete	bartlett_chi2	bartlett_p	kmo_overall
0	Combined	141	212.688137	1.135030e-33	0.647914

[46]:

```
eig_ses_combined = run_scree_analysis(df, "Combined", ses_efa_items)
eig_ses_combined
```



[46]:

	factor_number	eigenvalue
0	1	2.480968
1	2	1.272444
2	3	1.096462
3	4	0.819858
4	5	0.570590
5	6	0.497186
6	7	0.262491

[47]:
```
fa_ses_1, d_ses_1, loadings_ses_1, variance_ses_1 = fit_efa(
    df=df,
    sample="Combined",
    items=ses_efa_items,
    n_factors=1,
    rotation=None
)

plot_loading_heatmap(loadings_ses_1, title="SES EFA loadings (Combined)",␣
↪figsize=(20, 8))
```

- These items are moving together as one common dimension. Factor signs are arbitrary in EFA.

Conclusion

- Based on the EFA, SES can be treated as one broad underlying dimension, while AI literacy is better represented by two related but distinct components: conceptual understanding and ability/confidence.

The excluded SES-related variables can be retained outside the main SES factor and incorporated later as separate covariates, subgroup variables, or sensitivity checks.

- Refiting the simple linear model using the factor scores: It returns a disappointing result: the SES factor is not significantly associated with either AI factor, and the effect sizes are very small.

3 Mediators

3.1 Mediator pairs

Each mediator is measured by two different questions. We should check the consistency of these pairs before using them in mediation analyses.

[48]:
```
mediator_reliability = mediator_reliability_table(df, mediator_map)
mediator_reliability
```

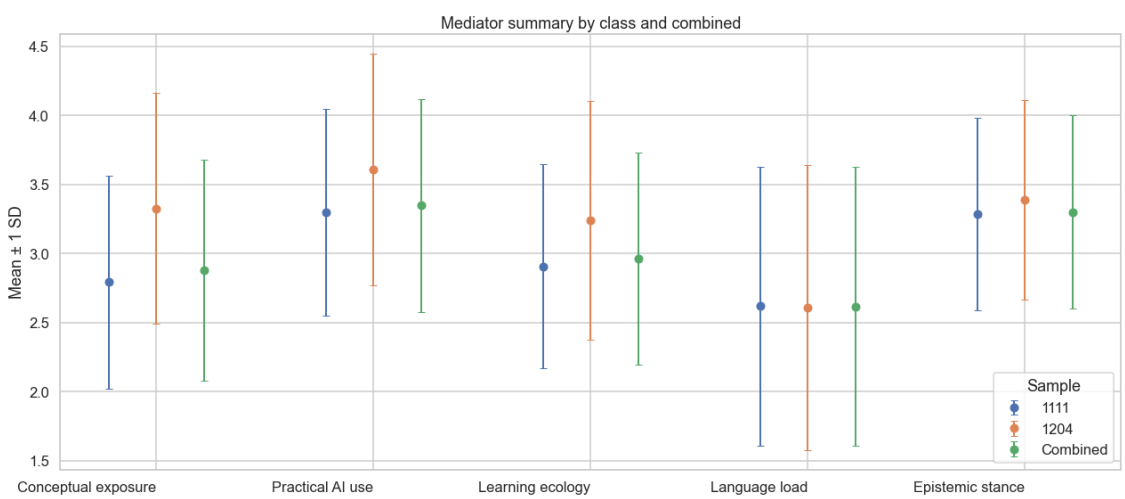
[48]:

	sample	mediator	n_complete	spearman_rho	alpha_2item
0	1111	conceptual_exposure	118	0.292140	0.486054
1	1111	practical_ai_use	118	0.223236	0.379432
2	1111	learning_ecology	118	0.147209	0.274513
3	1111	language_load	118	0.601679	0.759845
4	1111	epistemic_stance	118	0.320206	0.564601
5	1204	conceptual_exposure	23	0.392293	0.567944
6	1204	practical_ai_use	23	0.367887	0.528373
7	1204	learning_ecology	23	0.399625	0.622269
8	1204	language_load	23	0.625154	0.779258
9	1204	epistemic_stance	23	0.360713	0.406300
10	Combined	conceptual_exposure	141	0.322244	0.522657
11	Combined	practical_ai_use	141	0.256532	0.419903
12	Combined	learning_ecology	141	0.184561	0.350871
13	Combined	language_load	141	0.603460	0.761378
14	Combined	epistemic_stance	141	0.334422	0.539159

- rho >= 0.30 , alpha > 0.5 means acceptable for a 2-item pair
- Most pairs show at least a modest positive association (by rho and Alpha 2)
- The weakest pair is learning_ecology (rho 0.18, p 0.028). This is still positive, but the association is fairly small. That suggests the two items may not be capturing exactly the same thing, even if they belong to the same broad domain.

Add composite mediator scores (averaging two items)

[49]:
```
df_med = add_mediator_composites(df, mediator_map)
# mediators table
med_summary = mediator_summary_table(df_med)
plot_mediator_summary(med_summary, errorbar="sd")
```



4 Mediation Analysis

[50]:
```
# merge factor scores into mediator dataframe
df_analysis = df_med.copy()

scores = fa_combined.transform(d_combined)
df_factors = pd.DataFrame(
    scores,
    index=d_combined.index,
    columns=["ai_factor1_score", "ai_factor2_score"]
)

if "ai_factor1_score" in df_factors.columns and "ai_factor2_score" in␣
 ↪df_factors.columns:
    df_analysis.loc[df_factors.index, "ai_factor1_score"] =␣
 ↪df_factors["ai_factor1_score"]
    df_analysis.loc[df_factors.index, "ai_factor2_score"] =␣
 ↪df_factors["ai_factor2_score"]

# Check the data
# df_analysis[["course", "ses_index", "ai_factor1_score", "ai_factor2_score"]].
 ↪head()

mediator_vars = [
    "conceptual_exposure_score",
    "practical_ai_use_score",
    "learning_ecology_score",
    "language_load_score",
    "epistemic_stance_score",
]

outcome_vars = [
    "ai_factor1_score",   # conceptual understanding
    "ai_factor2_score",   # ability/confidence
]

mediation_results = run_all_simple_mediations(
    df_analysis,
    x="ses_index",
    mediators=mediator_vars,
    outcomes=outcome_vars,
    n_boot=3000,
    seed=2026
```

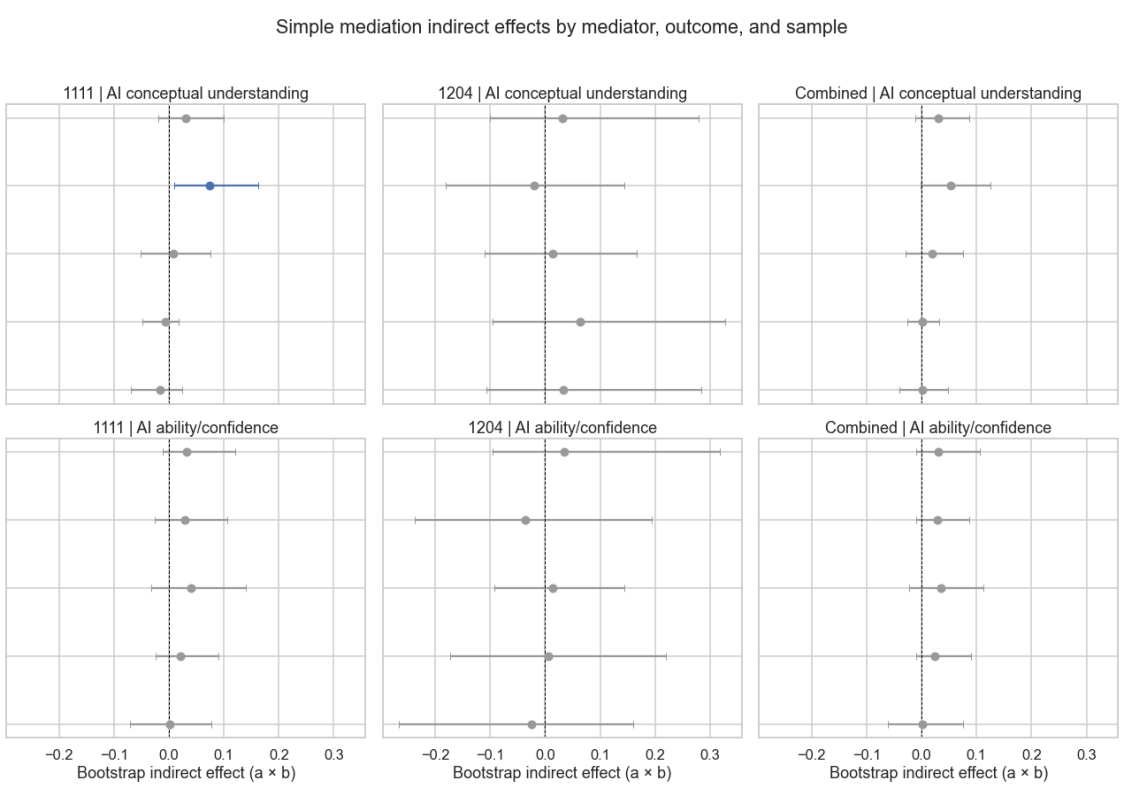
```
)

label_map = {
    "conceptual_exposure_score": "Conceptual exposure",
    "practical_ai_use_score": "Practical AI use",
    "learning_ecology_score": "Learning ecology",
    "language_load_score": "Language load",   # or Language ease if reversed
    "epistemic_stance_score": "Epistemic stance",
    "ai_factor1_score": "AI conceptual understanding",
    "ai_factor2_score": "AI ability/confidence",
}

mediation_results["mediator"] = mediation_results["mediator"].map(label_map)
mediation_results["outcome"] = mediation_results["outcome"].map(label_map)
```

4.0.1 The indirect-effect forest plot

- The a path is SES -> Mediator_i
- The b path is the Mediator_i -> AI literacy while controlling for SES.

Each point is the estimated indirect effect a × b, and the horizontal line is the bootstrap 95% confidence interval. If the interval stays on one side of zero, that mediator shows clearer evidence of an indirect pathway.

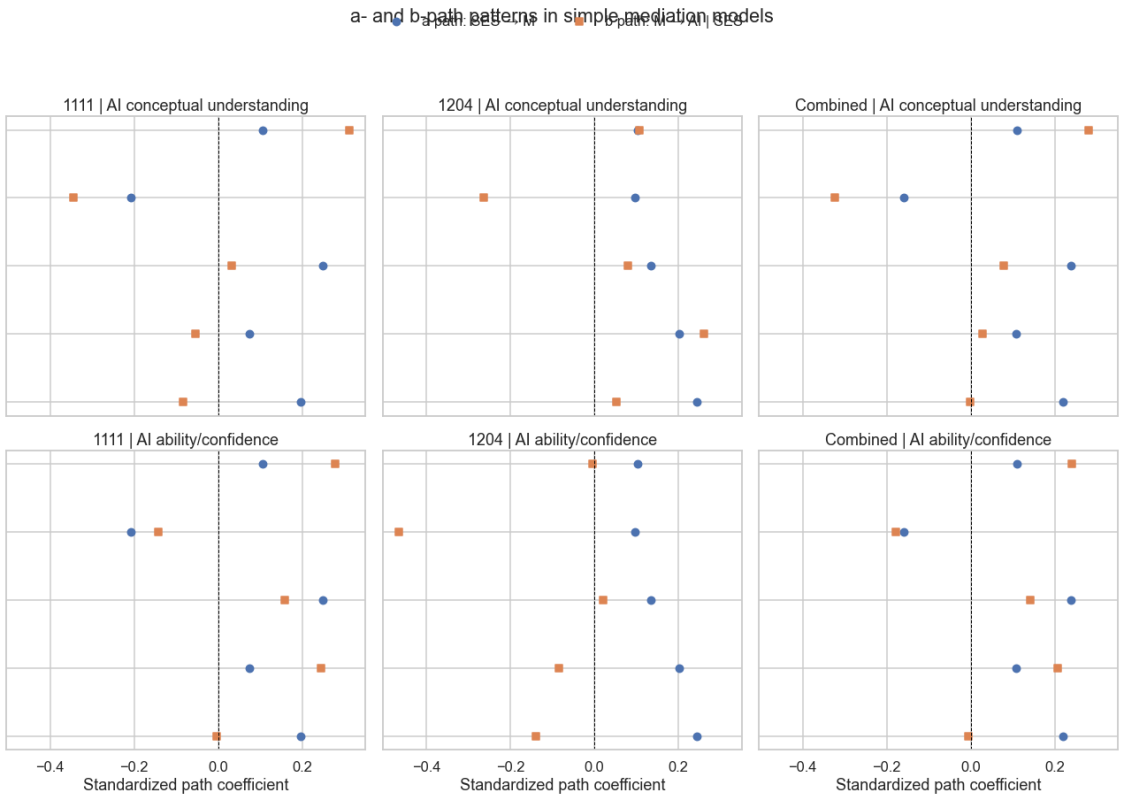
[51]:
```
plot_indirect_effect_forest(mediation_results)
```



4.0.2 a–b path plot

We need both paths to get a meaningful indirect effect, because mediation only exists when SES is linked to the mediator and the mediator is linked to AI literacy.

[74]:
```
plot_a_b_paths(mediation_results)
```



4.0.3 Main results

- 1204 is too unstable to support much interpretation because of the very small sample The combined sample and 1111 show more interpretable signals.
- The strongest mediation evidence is for Language load → AI conceptual understanding.
- A secondary, weaker candidate is Epistemic stance → AI conceptual understanding.
- For AI ability/confidence, none of the mediators shows a clearly convincing indirect effect.

Strong significant mediation:

1. Higher SES → Less language burden → Better conceptual AI understanding
2. Higher SES → Stronger epistemic stance → Better conceptual AI understanding

Expect but weak results

1. More SES → more practical AI use → more AI ability/confidence The evidence is weak, mainly because the a_path is not strong enough.
2. Conceptual exposure does not appear to be the mechanism linking SES to AI literacy in these simple mediation models.

---

5 Check 4 left-out SES categories

The following 4 variables are treated as non-ordinal SES indicators and are not included in the main SES factor: - Housing type - School type - Household size - Home language

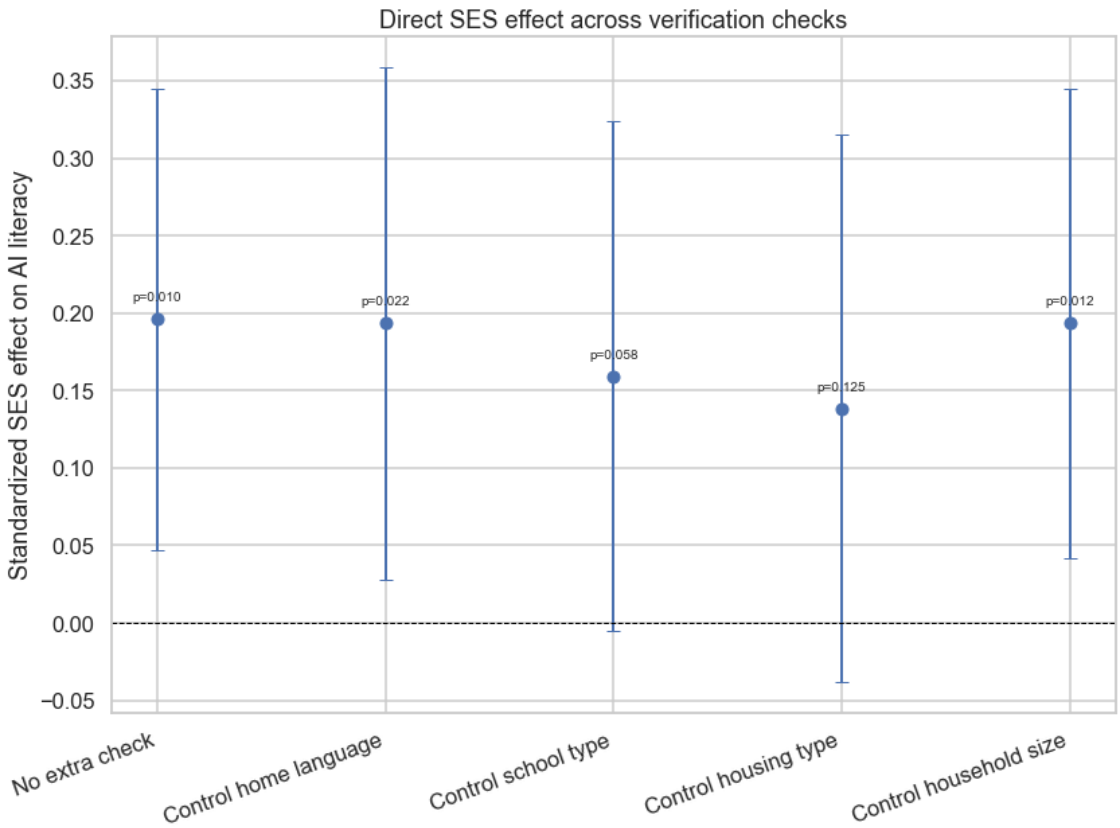
[53]:
```
check_groups = [None, "home_language","school", "housing", "household_size"]

direct_verify = pd.concat(
    [verify_direct_effect(df_analysis, check_group=g, sample="Combined") for g␣
 ↪in check_groups],
    ignore_index=True
)

med_verify = pd.concat(
    [verify_one_mediation(df_analysis, "language_load_score",␣
 ↪"ai_factor1_score", check_group=g, sample="Combined")
     for g in check_groups],
    ignore_index=True
)
```

5.1 Direct effect from SES -> AI literacy

The coefficient β_1 in AI literacy $= \beta_0 + \beta_1 \text{SES} + \varepsilon$: - about 0.19 with no extra check - about 0.19 after controlling for home language - about 0.19 after controlling for household size - shrink to 0.16 with school - shrink to 0.14 with housing

The main effect of SES -> AI literacy is fairly stable to home-language and household-size verification. Part of the SES effect overlaps with school and housing context.

[54]:
```
direct_verify
plot_direct_effect_verification(direct_verify)
```

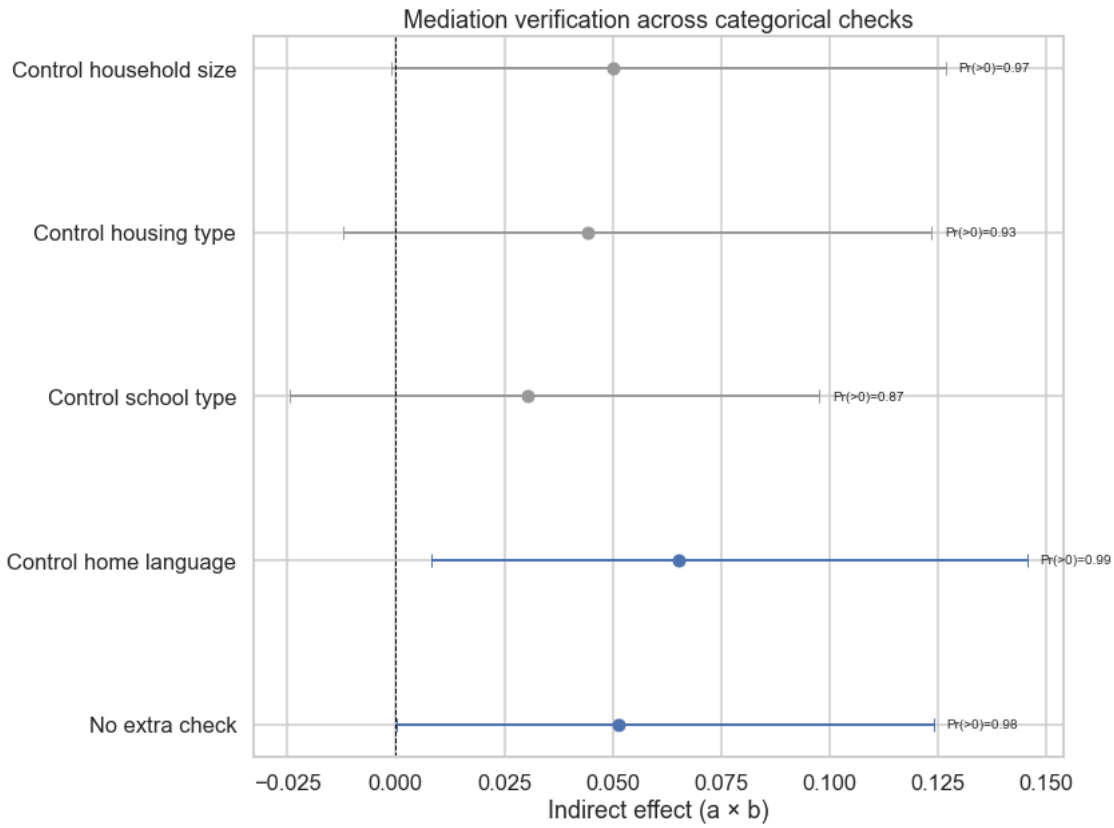


5.2 Mediating effects of SES on AI literacy through left-out SES categories

Strongest mediating effect:

SES → lower language load → better AI conceptual understanding

[55]:
```
med_verify
plot_mediation_verification(med_verify)
```



Result: 1. The indirect effect: SES -> language load -> AI conceptual understanding still positive

- a-path stays negative (higher SES -> lower language load)
- b-path stays negative (lower language load -> better conceptual AI understanding)

2. The indirect effect increases from about 0.051 to 0.065 controling for home language

- bootstrap CI stays above zero and proportion positive rises to about 0.99
- the language-load pathway is not simply a proxy for home-language background.

In plain language: • it is not just that some students speak different languages at home • it is that SES is still related to language burden, and language burden is still related to AI conceptual understanding, even after taking home-language background into account

---

6 Check 3 weaker mediators

- Practical AI use → AI ability/confidence looks like the most promising of the weak paths
- Learning ecology looks weak partly because the measurement is weak.

[56]:
```
diag_paths = diagnose_mediation_paths(mediation_results)
diag_paths[[
```

```
    "sample", "mediator", "outcome",
    "a_path", "a_p", "b_path", "b_p",
    "indirect_ab", "indirect_ci_low_95", "indirect_ci_high_95",
    "diagnosis"
]].round(3)
```

[56]:

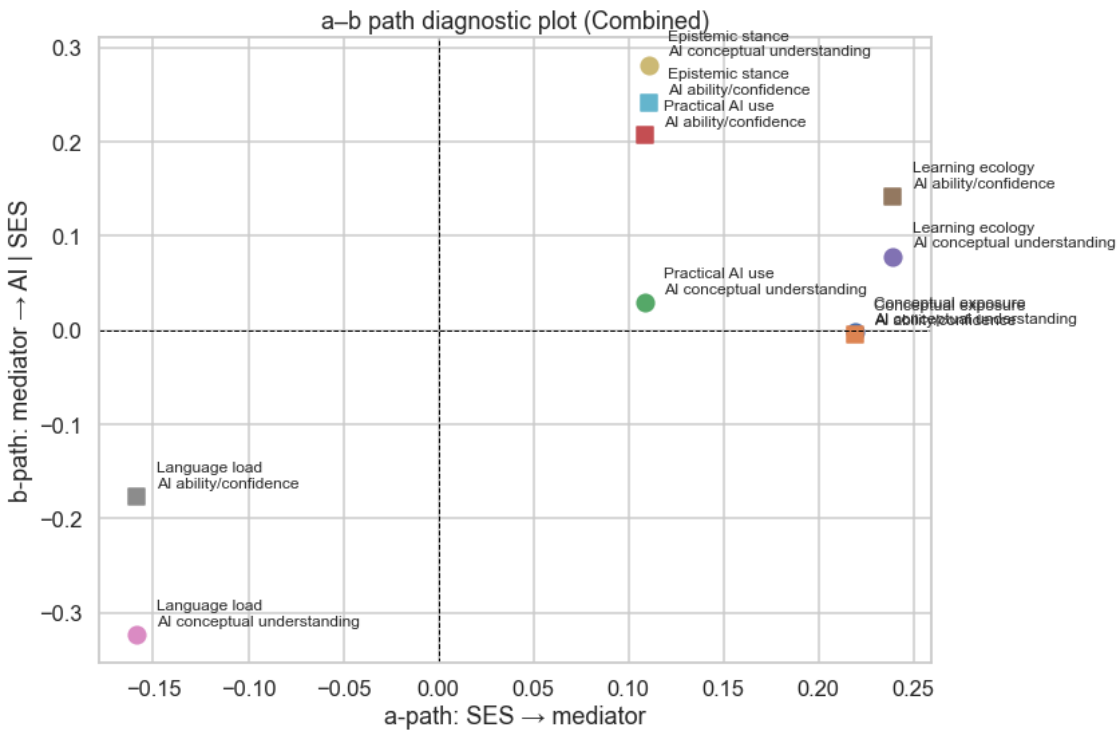
	sample	mediator	outcome	a_path	a_p	b_path	b_p	indirect_ab	indirect_ci_low_95	indirect_ci_high_95
0	1111	Conceptual exposure	AI conceptual understanding	0.197	0.031	-0.083	0.418	-0.016	-0.069	0.024
1	1111	Conceptual exposure	AI ability/confidence	0.197	0.031	-0.002	0.989	-0.000	-0.071	0.078
2	1111	Practical AI use	AI conceptual understanding	0.075	0.441	-0.055	0.623	-0.004	-0.048	0.017
3	1111	Practical AI use	AI ability/confidence	0.075	0.441	0.246	0.058	0.018	-0.024	0.090
4	1111	Learning ecology	AI conceptual understanding	0.250	0.005	0.033	0.776	0.008	-0.051	0.076
5	1111	Learning ecology	AI ability/confidence	0.250	0.005	0.159	0.310	0.040	-0.032	0.140
6	1111	Language load	AI conceptual understanding	-0.208	0.032	-0.345	0.002	0.072	0.010	0.163
7	1111	Language load	AI ability/confidence	-0.208	0.032	-0.143	0.311	0.030	-0.026	0.105
8	1111	Epistemic stance	AI conceptual understanding	0.107	0.209					
9	1111	Epistemic stance	AI ability/confidence	0.107	0.209					
10	1204	Conceptual exposure	AI conceptual understanding	0.243	0.252					
11	1204	Conceptual exposure	AI ability/confidence	0.243	0.252					
12	1204	Practical AI use	AI conceptual understanding	0.202	0.461					
13	1204	Practical AI use	AI ability/confidence	0.202	0.461					
14	1204	Learning ecology	AI conceptual understanding	0.135	0.529					
15	1204	Learning ecology	AI ability/confidence	0.135	0.529					
16	1204	Language load	AI conceptual understanding	0.097	0.653					
17	1204	Language load	AI ability/confidence	0.097	0.653					
18	1204	Epistemic stance	AI conceptual understanding	0.103	0.704					
19	1204	Epistemic stance	AI ability/confidence	0.103	0.704					
20	Combined	Conceptual exposure	AI conceptual understanding	0.219	0.006					
21	Combined	Conceptual exposure	AI ability/confidence	0.219	0.006					
22	Combined	Practical AI use	AI conceptual understanding	0.109	0.218					
23	Combined	Practical AI use	AI ability/confidence	0.109	0.218					
24	Combined	Learning ecology	AI conceptual understanding	0.239	0.002					
25	Combined	Learning ecology	AI ability/confidence	0.239	0.002					
26	Combined	Language load	AI conceptual understanding	-0.159	0.068					
27	Combined	Language load	AI ability/confidence	-0.159	0.068					
28	Combined	Epistemic stance	AI conceptual understanding	0.111	0.162					
29	Combined	Epistemic stance	AI ability/confidence	0.111	0.162					

8	0.312	0.015	0.033	-0.019	0.099
9	0.278	0.100	0.030	-0.011	0.121
10	0.052	0.852	0.013	-0.107	0.284
11	-0.139	0.672	-0.034	-0.266	0.160
12	0.261	0.357	0.053	-0.095	0.328
13	-0.084	0.790	-0.017	-0.173	0.220
14	0.080	0.791	0.011	-0.111	0.167
15	0.022	0.945	0.003	-0.093	0.145
16	-0.263	0.211	-0.025	-0.180	0.143
17	-0.465	0.026	-0.045	-0.238	0.193
18	0.106	0.701	0.011	-0.101	0.279
19	-0.005	0.989	-0.000	-0.095	0.317
20	-0.002	0.980	-0.001	-0.041	0.048
21	-0.005	0.975	-0.001	-0.062	0.076
22	0.029	0.763	0.003	-0.026	0.031
23	0.207	0.080	0.022	-0.009	0.091
24	0.078	0.448	0.019	-0.030	0.076
25	0.141	0.300	0.034	-0.022	0.112
26	-0.323	0.000	0.051	-0.001	0.125
27	-0.177	0.142	0.028	-0.009	0.087
28	0.281	0.008	0.031	-0.011	0.087
29	0.241	0.113	0.027	-0.010	0.106

	diagnosis
0	fails at b-path
1	fails at b-path
2	both paths weak
3	both paths weak
4	fails at b-path
5	fails at b-path
6	supported mediation
7	fails at b-path
8	fails at a-path
9	both paths weak
10	both paths weak
11	both paths weak
12	both paths weak
13	both paths weak
14	both paths weak
15	both paths weak
16	both paths weak
17	fails at a-path
18	both paths weak
19	both paths weak
20	fails at b-path
21	fails at b-path
22	both paths weak

```
23      both paths weak
24      fails at b-path
25      fails at b-path
26      fails at a-path
27      both paths weak
28      fails at a-path
29      both paths weak
```

[75]:
```
plot_ab_diagnostic(mediation_results, sample="Combined")
```



[67]:
```
compare_one_mediator_across_outcomes(mediation_results, "Practical AI use",␣
↪sample="Combined").round(3)
```

[67]:
```
      sample          mediator                      outcome  a_path    a_p  \
23  Combined  Practical AI use        AI ability/confidence   0.109  0.218
22  Combined  Practical AI use  AI conceptual understanding   0.109  0.218

    b_path    b_p  indirect_ab  indirect_ci_low_95  indirect_ci_high_95  \
23   0.207  0.080        0.022              -0.009                0.091
22   0.029  0.763        0.003              -0.026                0.031

    prop_ab_positive
23             0.890
```

```
22             0.584
```

[68]:
```
compare_one_mediator_across_outcomes(mediation_results, "Learning ecology",␣
↪sample="Combined").round(3)
```

[68]:
```
      sample          mediator                       outcome  a_path    a_p  \
25  Combined  Learning ecology        AI ability/confidence   0.239  0.002
24  Combined  Learning ecology  AI conceptual understanding   0.239  0.002

    b_path    b_p  indirect_ab  indirect_ci_low_95  indirect_ci_high_95  \
25   0.141  0.300        0.034              -0.022                0.112
24   0.078  0.448        0.019              -0.030                0.076

    prop_ab_positive
25             0.866
24             0.782
```

[69]:
```
compare_one_mediator_across_outcomes(mediation_results, "Epistemic stance",␣
↪sample="Combined").round(3)
```

[69]:
```
      sample          mediator                       outcome  a_path    a_p  \
29  Combined  Epistemic stance        AI ability/confidence   0.111  0.162
28  Combined  Epistemic stance  AI conceptual understanding   0.111  0.162

    b_path    b_p  indirect_ab  indirect_ci_low_95  indirect_ci_high_95  \
29   0.241  0.113        0.027              -0.010                0.106
28   0.281  0.008        0.031              -0.011                0.087

    prop_ab_positive
29             0.887
28             0.925
```

[ ]:
```

```